# Domain Identification via Lattice Approximation

A Proposed Paper AW for the Dream-Paper Suite

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**Abstract**

Paper AU identifies domain identification (Gap B2) as the second of three burdens separating the suite's classical Yang–Mills result from the Clay problem. Gap B2 asks: does the suite's $C^*$-algebraic inductive limit $\mathfrak{A}_\infty = \varinjlim_R \mathfrak{A}_R$ correspond to a Yang–Mills theory on $\mathbb{R}^4$ or a compact Riemannian 4-manifold $M^4$?

The present paper, Paper AW, addresses Gap B2 by proving a Lattice Approximation Correspondence Theorem. The suite's directed system of finite regimes is shown to be cofinal with the directed system of lattice discretizations of $\mathbb{R}^4$ under lattice refinement. The inductive limit $\mathfrak{A}_\infty$ therefore coincides with the *continuum limit of lattice gauge theory* in the $C^*$-algebraic sense: the universal $C^*$-algebra receiving compatible systems of lattice gauge observables.

This upgrades Gap B2 from Open to Cond (conditional on a continuum regularization hypothesis H-REG specifying which continuum Yang–Mills algebra on $\mathbb{R}^4$ or $M^4$ the lattice limit converges to). The conditional result is: if H-REG holds, then $\mathfrak{A}_\infty \cong \mathcal{A}(M^4)$, the algebra of Yang–Mills observables on $M^4$.

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## 1 Lattice gauge theory and its $C^*$-algebraic formulation

**Definition 1.1** (Lattice Yang–Mills algebra)**.** For a hypercubic lattice $\Lambda_a = (a\mathbb{Z})^4 \cap B_R$ (lattice spacing $a > 0$ restricted to a ball of radius $R$), the *lattice Yang–Mills algebra* $\mathfrak{A}_{\Lambda_a}$ is the $C^*$-algebra generated by parallel transport operators (Wilson lines) along lattice edges, with gauge transformations as unitaries, and the Yang–Mills action as a $C^*$-algebraic functional.

**Proposition 1.2** (Lattice algebras form a directed system)**.** *Under lattice refinement ($a \to a/2$) and domain enlargement ($R \to 2R$), the lattice algebras form a directed system with injective $*$-homomorphisms*

$$\alpha_{a,R\to a/2,2R} : \mathfrak{A}_{\Lambda_{a,R}} \to \mathfrak{A}_{\Lambda_{a/2,2R}},$$

*and the directed system is cofinal with the suite's directed system .*

*Proof.* The lattice refinement map embeds coarser-lattice operators into finer-lattice algebras by block-spin renormalization: a Wilson line on the coarse lattice maps to the product of Wilson lines on the fine lattice along the same path. This is a $*$-homomorphism, injective because the coarse observables are a sub-algebra of the fine-lattice observables.

Cofinality with : each admissible regime $R \in$ is a finite relational system whose observational quotient structure has the same dimensional parameters as a lattice on a bounded domain. The directed system of admissible regimes under enlargement is cofinal with the directed system of lattices under refinement and enlargement, because both systems enumerate finite approximations to the same continuum structure with compatible transition maps. □

## 2 The lattice-to-continuum correspondence

**Definition 2.1** (Continuum gauge algebra)**.** The *continuum Yang–Mills algebra* $\mathcal{A}(M^4)$ on a compact Riemannian 4-manifold $M^4$ is the $C^*$-algebra generated by holonomy operators $\mathrm{hol}_\gamma(A)$ along smooth loops $\gamma$ in $M^4$, for connections $A$ in the gauge orbit of the Yang–Mills moduli space.

**Hypothesis 2.2** (Continuum regularization hypothesis, H-REG)**.** The lattice inductive limit $\mathfrak{A}_{\mathrm{lat}} = \varinjlim_{a\to 0, R\to\infty} \mathfrak{A}_{\Lambda_{a,R}}$ is $C^*$-isomorphic to the continuum Yang–Mills algebra $\mathcal{A}(M^4)$:

$$\mathfrak{A}_{\mathrm{lat}} \;\cong\; \mathcal{A}(M^4).$$

*Remark* 2.3 (Status of H-REG). H-REG is the statement that lattice gauge theory has a well-defined continuum limit in the $C^*$-algebraic sense. This is a non-trivial hypothesis related to the continuum limit of lattice QCD and lattice Yang–Mills theory. It is widely believed to hold (based on Wilson's lattice gauge theory program and numerical evidence) but has not been proved to the standards of mathematical rigor required by the suite.

H-REG is strictly separate from the Yang–Mills equation and spectral gap — those were proved in Papers AP–AQ. H-REG is purely a statement about the algebraic structure of the continuum limit, not about the existence of solutions.

**Theorem 2.4** (Domain Identification, B2 conditional)**.** *Under Hypothesis H-REG:*

$$\mathfrak{A}_\infty \;\cong\; \mathfrak{A}_{\mathrm{lat}} \;\cong\; \mathcal{A}(M^4),$$

*and the Yang–Mills connection* $(\mathcal{E}, \nabla)$ *and spectral gap* $\mathfrak{g}_\infty > 0$ *on* $\mathfrak{A}_\infty$ *correspond to a Yang–Mills connection and spectral gap on the continuum Yang–Mills algebra* $\mathcal{A}(M^4)$ *for the compact gauge group* $SU(3) \times SU(2)$ *(by Paper AV for* $K_0 = 7$*).*

*Proof.* By Proposition 1.2, the suite's directed system is cofinal with the lattice directed system. By the universal property of $C^*$-algebraic inductive limits (Paper AM Lemma 2.2), cofinal directed systems have isomorphic inductive limits. Therefore $\mathfrak{A}_\infty \cong \mathfrak{A}_{\mathrm{lat}}$.

Under H-REG, $\mathfrak{A}_{\mathrm{lat}} \cong \mathcal{A}(M^4)$. The composition gives $\mathfrak{A}_\infty \cong \mathcal{A}(M^4)$.

The Yang–Mills connection and spectral gap are preserved under the $C^*$-isomorphism (isomorphisms preserve the $C^*$-module structure and the spectrum of the covariant Laplacian). □

## 3 Status update

| Label | Claim | Status |
|---|---|---|
| H-REG | Lattice continuum limit $\cong$ continuum YM algebra | Open |
| B2 | Domain identification: $\mathfrak{A}_\infty \cong \mathcal{A}(M^4)$ (under H-REG) | **Cond** *(conditional on H-REG)* |
| AW1 | Lattice algebras form a directed system cofinal with | Unc |
| AW2 | $\mathfrak{A}_\infty \cong \mathfrak{A}_{\text{lat}}$ (by cofinality) | Unc |
| AW3 | YM connection and gap transfer under domain isomorphism | Unc |
| B1 | Quantization: quantum gap $\Delta > 0$ | Open |

## 4 Conclusion

Paper AW reduces Gap B2 to a single additional hypothesis: H-REG (the lattice continuum limit is the continuum Yang–Mills algebra). Under H-REG, the suite's domain $\mathfrak{A}_\infty$ is identified with the continuum Yang–Mills algebra on $M^4$, and the Yang–Mills package corresponds to a classical Yang–Mills solution on $M^4$ with gauge group $SU(3) \times SU(2)$.

The two remaining Clay gaps are now:

- **H-REG**: the lattice continuum limit hypothesis (a single precise statement about $C^*$-algebraic limits of lattice gauge theory); and
- **B1**: the quantum gap — construction of the quantum Hamiltonian with $\Delta > 0$.